Nonlinear optics parameters for 350 nm silicon nitride platform. **SGM 16/1/25**

We want to establish the nonlinear interaction coefficient, dispersion and prospective finesse of ring resonators built using the AN350 SiN platform. This lets us calculate the normalized driving power *X* for Kerr comb operation.

Dispersion:

The dispersion of 350 nm thick SiN overclad with silica can be inferred from the following two graphs of Refs. [1] and [2] respectively. Nb: [2] is with silicon-rich SiN

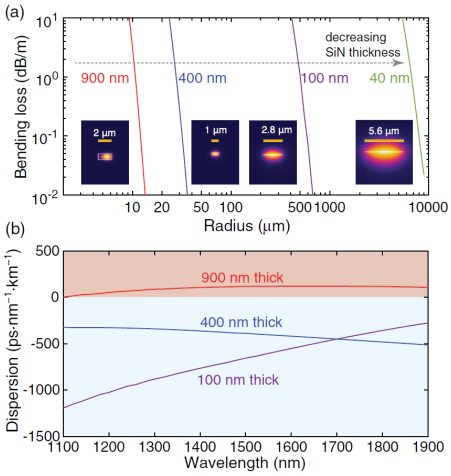
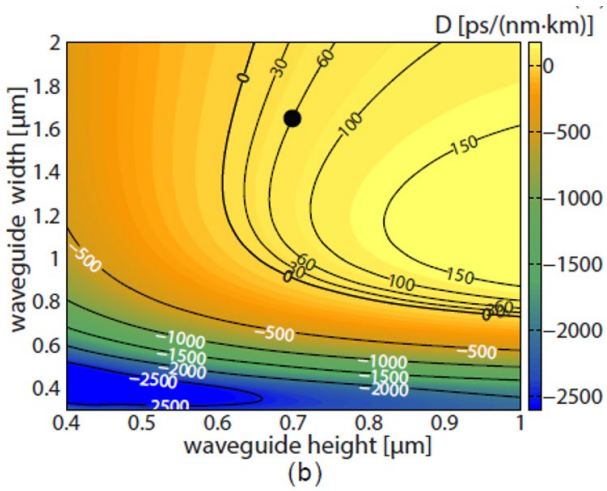


Fig. 2. (a) SiN waveguide bending loss versus bending radius of the TE0 optical mode for four different SiN core thicknesses. Insets show the mode profiles for each waveguide geometry with negligible bending loss at 20 µm, 50 µm, 1 mm, and 10 mm respectively. (b) Waveguide dispersion for waveguides with SiN core thickness and width selected from (a).

[1]



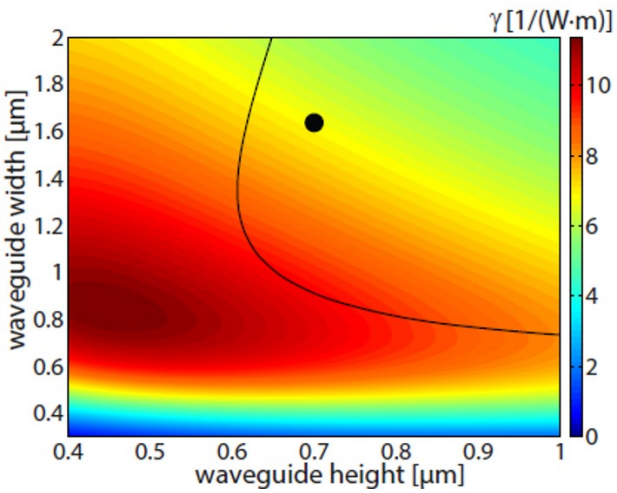
[2]

So for our situation: T = 350 nm, W = 2000 nm, and λ = 1550 nm

It is fair to assume D = -500 ps/(nm.km)

Or equivalently β_2 = +392 ps^2/km i.e. Strong normal dispersion

Nonlinearity:



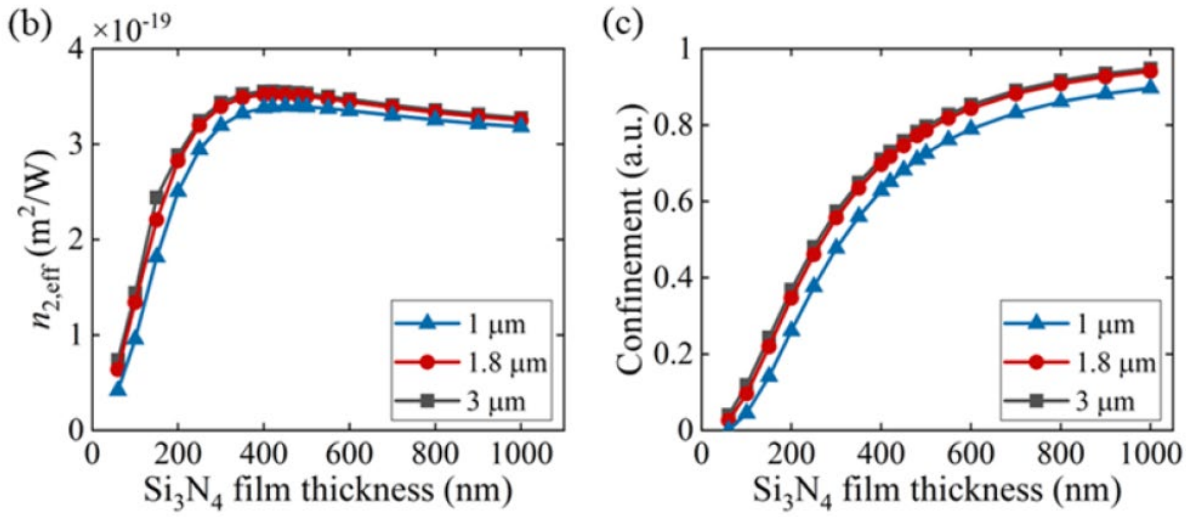
From [2]

Silicon-rich SiN has an n_2 5 times SiN [2]

So for our situation: T = 350 nm, W = 2000 nm, and λ = 1550 nm

We get γ = 1.4 $W^{-1}m^{-1}$ (so one thousand times that of SMF28)

Alternatively, from Ref [3] we have for T = 400 nm



So for our situation: T = 350 nm, W = 2000 nm, and λ = 1550 nm

n2eff ~ 3 x 10^{-19} m^2/W and confinement ~ 0.5 so Aeff ~ 1.4 $(um)^2$

and so γ = 1.2 $W^{-1}m^{-1}$

Resonator finesse:

From Ligentec PDK:

$\alpha_{dB} = 7.5 \text{ dB/m}$ (for a 1.7 um waveguide at 1550 nm)

Assume Diameter = 400 um, n = 2, λ = 1550 nm

$F = \frac{\pi}{\alpha} = \frac{2 \cdot 4.343 \cdot \pi}{\alpha_{dB} \pi D}$ (assuming no outcoupling), or ,

$F_{cc} = \frac{4.343}{\alpha_{dB} D}$ = 1440 (for a critically coupled resonator)

Normalised driving power:

$$X = P_{in}\frac{\gamma L\theta}{\alpha^3}, \text{ and } \frac{X_{cc}}{P_{in}} = \frac{\gamma L F^2}{\pi^2}.$$

γ = 1.2 $W^{-1}m^{-1}$, D = 400 um, L = 1.25 mm, F = 1440

$$\frac{X_{cc}}{P_{in}} = 0.315\ (\mathrm{mW}^{-1})$$

This is not bad, should be able to reach parametric oscillation at about ~3 mW and combs at ~ 20 mW on chip power into the bus …

References:

[1] Xiang C, Jin W, Bowers JE. Silicon nitride passive and active photonic integrated circuits: trends and prospects. Photonics research. 2022 May 20;10(6):A82-96.

[2] Krückel CJ, Fülöp A, Klintberg T, Bengtsson J, Andrekson PA, Torres-Company V. Linear and nonlinear characterization of low-stress high-confinement silicon-rich nitride waveguides. Optics express. 2015 Sep 23;23(20):25827-37.

[3] Zhang Y, Zhang S, Bi T, Del'Haye P. Geometry optimization for dark soliton combs in thin multimode silicon nitride microresonators. Optics Express. 2023 Nov 22;31(25):41420-7.